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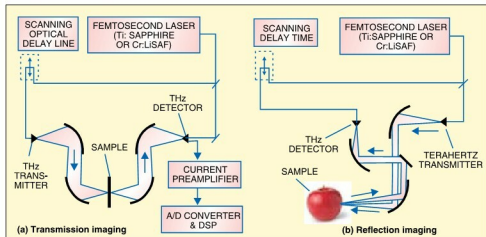


Fig. 3: Experimental setups used for (a) transmission and (b) reflection imaging

phase of the THz wave pulse, therefore providing the real and imaginary parts of the dielectric constant without relying on the Kramers-Kronig relations of ordinary spectroscopy. This allows precise measurement of the refractive index and absorption coefficients of samples that interact with the THz waves. The absorption spectrum and refractive index are calculated for every pixel.

Electro-optic crystals have been widely used in the THz-TDS and related imaging applications. In a conventional experimental setup of the THz-TDS, separate THz transmitter and THz receiver are used for generation and detection of the THz signal.

Tomography is imaging by sectioning. It is a THz technique that shows a single plane of the object under examination, typically a part of the human body.

The three-dimensional THz tomographic imaging can be taken by different methods as follows: THz diffraction tomography (THz DT), THz computed tomography (THz CT), THz binary lens tomography and THz digital holography. THz diffraction tomography uses the THz wave as a probe beam to interact with a target, and then reconstructs the three-dimensional image of the target using the THz waves scattered by the target. THz CT is based on geometrical optics and inspired by X-ray CT. THz binary lens tomography uses the frequency-dependent focal length

property of binary lenses to obtain tomographic images of an object. By separating the multiple scattered THz waves of different scattering orders, the digital holography method is used to reconstruct the sparsely distributed scattering centres.

The interaction between a coherent THz pulse and an object provides rich information about the object under study. Therefore three-dimensional THz imaging is a very useful tool for inspecting or characterising the objects. For example, three-dimensional THz imaging can be used to detect and identify defects inside a space shuttle's insulation tile.

The technical infrastructure is common even in security and bio-sensing systems. For example, the combination of a small light source with relatively high power output and a large-format sensitive array detector would significantly decrease the cost of THz imaging system for use in security.

In particular, many materials have distinct absorptive and dispersive properties in this spectral range, so THz imaging could reveal interesting features. For example, it would be possible to image the distribution of specific proteins or water in tissue, or buried metal layers in semiconductors; the present source would allow full-field, real-time capture of such images.

High-peak- and average-power THz sources are also critical in driv-